Math toolbox

Geometric: $\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}$ $(|a|<1)$, $\sum_{k=0}^{N-1} a^k = \frac{1-a^N}{1-a}$, $\sum_{k\ge 0} k a^k = \frac{a}{(1-a)^2}$, $\sum_{k\ge 1} a^k = \frac{a}{1-a}$.

Complex: $z = a+jb = re^{j\theta}$, $r=\sqrt{a^2+b^2}$, $\theta = \operatorname{atan2}(b,a)$; $e^{j\theta} = \cos\theta + j\sin\theta$; $\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$; $\frac{1}{z} = \frac{\bar z}{|z|^2}$; $|z_1 z_2| = |z_1||z_2|$, $\angle(z_1 z_2) = \angle z_1 + \angle z_2$; $e^{j\pi} = -1$, $e^{j\pi/2} = j$; angles mod 2π.

2×2: $A = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$: $A^{-1} = \frac{1}{ad-bc}\left[\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right]$, $\det(zI-A) = z^2 - \operatorname{tr}(A) z + \det A$, $\lambda_{1,2} = \frac{\operatorname{tr}}{2} \pm \sqrt{\frac{\operatorname{tr}^2}{4} - \det}$. Triangular ⇒ eigenvalues on diagonal.

Taylor: $e^x = \sum \frac{x^k}{k!}$, $\sin x \approx x$, $\cos x \approx 1 - \frac{x^2}{2}$, $\sqrt{q} \approx \sqrt{q_e} + \frac{\Delta q}{2\sqrt{q_e}}$.

W1 Signals & system properties

Signals CT $x(t), t\in\mathbb{R}$; DT $x[n], n\in\mathbb{Z}$ (sequence). $\{x[n]\}$ = whole signal. Arrow ↑ marks $n=0$; unlisted samples $=0$. $\{2,\underset{\uparrow}{3},1\}$: $x[-1]=2$, $x[0]=3$, $x[1]=1$.

Operations (compose in any order) $\alpha x[n]$ scaling; $x[n-k]$ $(k>0)$ **delay** = shift **right**; $x[n+k]$ **advance** = shift **left**; $x[-n]$ reversal about $n=0$. Locate sample: set argument = index, e.g. $x[n-2]$ has $x[0]$ at $n=2$. $x[-n+k] = x[-(n-k)]$: *first* flip, *then* delay by k.

Key signals $\delta[n]=1$ for $n=0$, else 0; $s[n]=1$ for $n\ge 0$, else 0. $s[n] = \sum_{k=-\infty}^{n} \delta[k]$, $\delta[n] = s[n] - s[n-1]$ $x[n]\delta[n-k] = x[k]\delta[n-k]$ (sifting)

Impulse decomposition: $x[n] = \sum_k x[k]\delta[n-k]$ e.g. $\{\underset{\uparrow}{1},2\} = \delta[n] + 2\delta[n-1]$.

Causal signal: $x[n]=0\ \forall n<0$ ($\neq$ causal system!).

Bounded: $\exists M<\infty$: $|x[n]|\le M\ \forall n$.

System $y = Gu$. Properties (prove with the definition; disprove with *one* counterexample):

Memoryl.	$y[n]$ depends only on $u[n]$
Causal	$y[n]$ depends only on $u[k], k\le n$
Linear	$G\{\alpha_1 u_1 + \alpha_2 u_2\} = \alpha_1 y_1 + \alpha_2 y_2$ $\forall u_i, \alpha_i$
TI	$u_1[n-k] \mapsto y_1[n-k]\ \forall k$
BIBO	$|u[n]|\le 1\ \forall n \Rightarrow |y[n]| \le M < \infty$

TI test: compute $y_2[n] = G\{u_1[n-k]\}$ and compare with $y_1[n-k]$ (replace *every* n by $n-k$).

Quick tells:
- Linear ⇒ $0\mapsto 0$. $u+2$, u^2, $\cos u$, $u[n]u[n-1]$: *not* linear (affine ≠ linear).
- Explicit n ($n\,u[n]$, $s[n]u[n]$), $u[2n]$, $u[-n]$: *not* TI. $g[n]u[n]$: linear $\forall g$.
- $u[n+k]$, $u[-n]$: *not* causal. $\cos u$: memoryless, stable, nonlinear.
- $n\,u[n]$ unstable ($u=s \Rightarrow y = ns[n]$); $y[n]=ay[n-1]+u[n]$ stable iff $|a|<1$.

W2 Impulse response & convolution

$h[n] := G\{\delta[n]\}$. For **LTI**: $\delta[n-k]\mapsto h[n-k]$ (TI), $c\delta[n-k]\mapsto ch[n-k]$ (L) ⇒ h determines G completely.

Convolution (needs L *and* TI): $y[n] = \sum_k u[k]h[n-k] = \sum_k h[k]u[n-k]$

By hand: flip h, slide, multiply overlap, sum; or $y[n] = \sum_{i+j=n} u[i]h[j]$ (polynomial product).

Support: lengths $L_y = L_u + L_h - 1$; start index $n_u + n_h$, end index $e_u + e_h$.

Check: $\sum_n y[n] = \left(\sum u\right)\left(\sum h\right)$.

Ex: $\{\underset{\uparrow}{1},2,3\} * \{\underset{\uparrow}{1},\frac{1}{2}\} = \{\underset{\uparrow}{1},\frac{5}{2},4,\frac{3}{2}\}$; $\{\underset{\uparrow}{1},-1,2\} * \{\underset{\uparrow}{1},1,1\} = \{\underset{\uparrow}{1},0,2,1,2\}$.

Properties commutative $u*h = h*u$; associative $(u*h_1)*h_2 = u*(h_1*h_2)$; distributive $u*(h_1+h_2) = u*h_1 + u*h_2$; identity $u*\delta = u$; shift $u*\delta[n-k] = u[n-k]$; ! $u*h \neq u\cdot h$.

Interconnections: series ⇒ h_1*h_2 (order irrelevant for LTI); parallel ⇒ h_1+h_2; direct path ⇒ $+\delta$. $\delta + \{\underset{\uparrow}{1},-1\} * \{\underset{\uparrow}{1},1\} = \{\underset{\uparrow}{2},0,-1\}$.

Deconvolution ($h[0]\neq 0$): solve recursively, $u[n] = \left(y[n] - \sum_{k\ge 1} h[k]u[n-k]\right)/h[0]$.

Step response $r[n] = G\{s[n]\} = \sum_{k=-\infty}^{n} h[k]$, $h[n] = r[n] - r[n-1]$. Steady state (stable): $r[\infty] = \sum_k h[k] = H(1)$.

Criteria for LTI **Causal** ⟺ $h[n]=0\ \forall n<0$. **Memoryless** ⟺ $h[n] = c\delta[n]$.

BIBO stable ⟺ $\sum_k |h[k]| < \infty$ (then $|y| \le \sum|h| \cdot \max|u|$).
- $\alpha^n s[n]$: $\sum = \frac{1}{1-|\alpha|}$, stable iff $|\alpha|<1$.
- $s[n]$: causal, bounded, *unstable* ($u=s \Rightarrow y = (n+1)s[n]$).
- $\frac{1}{n+1}s[n]$: $h\to 0$ but harmonic sum diverges ⇒ unstable. Decay/boundedness are *not* enough.
- 2^n for $n\le 0$: $\sum = 2$ ⇒ stable, not causal (support matters, not base).
- Cascade $s*(\delta[n]-\delta[n-1]) = \delta$: unstable subsystem can be cancelled (pole-zero cancellation).

FIR (causal): $h[n]=0\ \forall n\ge N$ ⇒ always stable. **IIR**: infinitely many nonzero $h[n]$ (stable or not).

W3 System representations

$h[n]$	convolution	analysis
LCCDE	recursion	real-time (DSP)
(A,B,C,D)	state space	simulation, design

LCCDE (LTI by def.; order N): $\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k u[n-k]$, $a_0 \neq 0$. Causal recursion: $y[n] = \frac{1}{a_0}\left(\sum_{k=0}^{M} b_k u[n-k] - \sum_{k=1}^{N} a_k y[n-k]\right)$.

LCCDE→ h: iterate with $u=\delta$, $y[n]=0$ for $n<0$. E.g. $y[n] - \frac{3}{4}y[n-1] = 2u[n] \Rightarrow h = 2(\frac{3}{4})^n s[n]$.

! Sign: $y[n] - a\,y[n-1] \Rightarrow$ ratio $+a$.

$h \to$ **LCCDE:** FIR: read off, $b_k = h[k]$. Sum of geometric terms: via $H(z)$ (W5).

State space (DT)

$q[n+1] = Aq[n] + Bu[n]$, $y[n] = Cq[n] + Du[n]$

$A_{N\times N}$ dynamics, $B_{N\times 1}$ input, $C_{1\times N}$ output, D feedthrough ($u[n]\to y[n]$ directly). State = memory: everything from the past that still affects the future; N = order (one per energy store/stored past value; inputs are *not* states).

SS → h $(q[0]=0,\ u=\delta;\ q[n] = A^{n-1}B)$

$$h = \{\underset{\uparrow}{D}, CB, CAB, CA^2B, \dots\}$$

i.e. $h[0]=D$, $h[n] = CA^{n-1}B$ for $n\ge 1$.

! $h[0]=D$, not CB. $D=0$ ⇒ response 1 sample late.

Ex: $(\frac{1}{3},1,2,1) \Rightarrow h = \{\underset{\uparrow}{1},2,\frac{2}{3},\frac{2}{9},\dots\}$.

With initial state: $y[n] = \underbrace{CA^n q[0]}_{\text{free}} + \underbrace{(h*u)[n]}_{\text{forced}}$.

LCCDE → SS (companion form, $a_0=1$) *Course form* (RHS $= b_0 u[n]$ only), $q[n] = [y[n-N], \dots, y[n-1]]^\top$:

$$A = \begin{bmatrix} 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \\ -a_N & -a_{N-1} & \cdots & -a_1 \end{bmatrix},\quad B = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix},$$

$C = [-a_N \cdots -a_1]$, $D = b_0$.

General (delayed inputs, $M\le N$, pad $b_k=0$): same A, $B = [0\cdots 0\ 1]^\top$, $C = [b_N - a_N b_0,\ \dots,\ b_1 - a_1 b_0]$, $D = b_0$.

1st order: $y[n] + a_1 y[n-1] = b_0 u[n] + b_1 u[n-1]$: $(A,B,C,D) = (-a_1, 1, b_1 - a_1 b_0, b_0)$.

Or choose state $q[n] = [y[n-1], u[n-1]]^\top$ (store past inputs too).

SS → LCCDE Eliminate q (write equation one step earlier, subtract) or via $H(z) = C(zI-A)^{-1}B + D$. Scalar: $y[n] - A\,y[n-1] = D\,u[n] + (CB - AD)\,u[n-1]$.

Ex: $(\frac{1}{4},1,\frac{3}{4},1)$: $y[n] - \frac{1}{4}y[n-1] = u[n] + \frac{1}{2}u[n-1]$.

Non-uniqueness $q' = Tq$ (T invertible): $(TAT^{-1}, TB, CT^{-1}, D)$ gives the same h. Compare realizations via h (or $H(z)$), *never* entrywise. E.g. $(\frac{1}{2},1,\frac{1}{2},1) \equiv (\frac{1}{2},2,\frac{1}{4},1)$.

CT state space

$$\dot q = A_c q + B_c u, \quad y = C_c q + D_c u$$

Build: first rows = definitions ($\dot q_1 = q_2$), last row = law solved for highest derivative.

MSD $m\ddot p + c\dot p + kp = F$, $q = [p, \dot p]^\top$: $A_c = \left[\begin{smallmatrix} 0 & 1 \\ -k/m & -c/m \end{smallmatrix}\right]$, $B_c = \left[\begin{smallmatrix} 0 \\ 1/m \end{smallmatrix}\right]$, $C_c = [1\ 0]$ (pos.) or $[0\ 1]$ (vel.), $D_c = 0$.

RC: $RC\dot y + y = u$: $A_c = -\frac{1}{RC}$, $B_c = \frac{1}{RC}$.

RLC $q = [v_C, i]^\top$: $A_c = \left[\begin{smallmatrix} 0 & 1/C \\ -1/L & -R/L \end{smallmatrix}\right]$, $B_c = \left[\begin{smallmatrix} 0 \\ 1/L \end{smallmatrix}\right]$.

Linearization Nonlinear $\dot q = f(q,u)$, $y = h(q,u)$.

1. **Equilibrium** (q_e, u_e): $0 = f(q_e, u_e)$, $y_e = h(q_e, u_e)$ (input need not be 0; may be several).

2. **Jacobians at equilibrium** (row i = eq. i, column j = var. j): $A_c = \frac{\partial f}{\partial q}\big|_e$, $B_c = \frac{\partial f}{\partial u}\big|_e$, $C_c = \frac{\partial h}{\partial q}\big|_e$, $D_c = \frac{\partial h}{\partial u}\big|_e$

3. **Deviations** $\Delta q = q - q_e$, $\Delta u = u - u_e$, $\Delta y = y - y_e$: $\Delta\dot q = A_c \Delta q + B_c \Delta u$, $\Delta y = C_c \Delta q + D_c \Delta u$.

LTI (constant matrices); valid only near equilibrium. ! No q, $\sin q$ left inside matrices.

Pendulum (θ from upright): $\dot q = \left[\begin{smallmatrix} x_2 \\ \frac{g}{L}\sin x_1 - c x_2 + \frac{1}{mL^2}u \end{smallmatrix}\right]$.

$u_e = -mgL\sin\theta_0$; $A_c = \left[\begin{smallmatrix} 0 & 1 \\ \frac{g}{L}\cos\theta_0 & -c \end{smallmatrix}\right]$, $B_c = \left[\begin{smallmatrix} 0 \\ \frac{1}{mL^2} \end{smallmatrix}\right]$. Upright $\theta_0=0$: $+\frac{g}{L}$; hanging $\theta_0=\pi$: $-\frac{g}{L}$.

Tank $\dot q = u - 2\sqrt{q}$, $u_e = 4$: $q_e = 4$, $\Delta\dot q = -\frac{1}{2}\Delta q + \Delta u$.

Drag $\dot v = u - v^2$: equilibria $(0,0)$, $(10,100)$, …; $A_c = -2v_e$.

W4 Sampling & discretization

Sampler S (CT→DT): $x[n] = x(nT_s)$, $f_s = \frac{1}{T_s}$.

ZOH H (DT→CT): $x(t) = x[n]$ for $nT_s \le t < (n+1)T_s$ (staircase, not interpolation).

$S\circ H = \mathrm{id}$ (hold then sample); $H\circ S \neq \mathrm{id}$ (inter-sample info lost).

Sampling can lose info: $\cos(2\pi t)$, $T_s=1 \Rightarrow x[n]=1$ (same as const.). $T_s=\frac{1}{2}$: $(-1)^n$; $T_s=\frac{1}{4}$: $\{1,0,-1,0\}$; $T_s=\frac{1}{3}$: $\{1,-\frac{1}{2},-\frac{1}{2}\}$.

DT freq. $\Omega = \omega T_s$.

Hold-and-sample architecture



$G_d = S G_c H$ (combination read right-to-left). $u[n], y[n]$ DT; $u(t), y(t)$ CT. Pure delay plant $y(t) = u(t-\tau)$, $kT_s < \tau < (k+1)T_s \Rightarrow y[n] = u[n-k-1]$ (rounded *up*); $\tau = 0 \Rightarrow y[n] = u[n]$.

Euler forward $\dot q \approx \frac{q(t+T_s) - q(t)}{T_s}$

$$A_d = I + T_s A_c, \quad B_d = T_s B_c$$

$C_d = C_c$, $D_d = D_c$ (no derivative in output eq.). Error $\mathcal{O}(T_s^2)$/step. **Stability:** need $|1 + T_s\lambda_i| < 1\ \forall$ eig. λ_i of A_c. Scalar $a<0$: stable iff $T_s < \frac{2}{|a|}$; monotone iff $T_s < \frac{1}{|a|}$; $\frac{1}{|a|} < T_s < \frac{2}{|a|}$: decays but alternates sign (fake oscillation). Ex: $\dot q = -4q + u$, $T_s = 0.2$: $a_d = 0.2$, $b_d = 0.2$.

! Common errors: forgetting I, sign, discretizing C, D.

Exact (ZOH) discretization Input constant on each interval ⇒ augment $r = [q; u]$:

$$M = \begin{bmatrix} A_c & B_c \\ 0 & 0 \end{bmatrix}, \quad e^{MT_s} = \begin{bmatrix} A_d & B_d \\ 0 & I \end{bmatrix}$$

$A_d = e^{A_c T_s}$ (top-left), B_d = top-right block, $C_d = C_c$, $D_d = D_c$. Exact at sampling instants.

Equivalently $B_d = \int_0^{T_s} e^{A_c\tau}\,d\tau\, B_c = A_c^{-1}(A_d - I)B_c$ (A_c invertible).

Scalar: $a_d = e^{aT_s}$, $b_d = \frac{b}{a}\left(e^{aT_s} - 1\right)$ ($a=0$: $b_d = bT_s$).

Eigenvalues: $\lambda \mapsto e^{\lambda T_s}$, $|e^{\lambda T_s}| = e^{\operatorname{Re}\lambda T_s}$ ⇒ CT stable ⇒ DT stable for *every* T_s. Stable 1st order: $0 < a_d < 1$.

	Euler	Exact
A_d	$I + T_s A_c$	$e^{A_c T_s}$
B_d	$T_s B_c$	top-right of e^{MT_s}
accuracy	$\mathcal{O}(T_s^2)$	exact (ZOH)
stability	small T_s only	always kept

Fast dynamics / large T_s ⇒ exact; slow & small T_s ⇒ Euler OK.

Matrix exponential by hand

$e^{At} = \sum_k \frac{(At)^k}{k!}$, $\frac{d}{dt}e^{At} = Ae^{At}$, $e^0 = I$.
- Diagonal: $e^{\operatorname{diag}(\lambda_i)t} = \operatorname{diag}(e^{\lambda_i t})$ (off-diag stays 0).
- Nilpotent ($M^k = 0$): series ends, e.g. $M^3 = 0$ for cart.
- Diagonalizable $A = V\Lambda V^{-1}$: $e^{At} = V e^{\Lambda t} V^{-1}$.
- $\left[\begin{smallmatrix} \lambda & 1 \\ 0 & \lambda \end{smallmatrix}\right]$: $e^{\lambda t}\left[\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right]$; $\left[\begin{smallmatrix} \sigma & \omega \\ -\omega & \sigma \end{smallmatrix}\right]$: $e^{\sigma t}\left[\begin{smallmatrix} \cos\omega t & \sin\omega t \\ -\sin\omega t & \cos\omega t \end{smallmatrix}\right]$
- Scalar block pattern: $M^k = \left[\begin{smallmatrix} a^k & a^{k-1}b \\ 0 & a^k \end{smallmatrix}\right]$.

Standard results:
- Integrator/tank $\dot q = u$: $a_d = 1$, $b_d = T_s$.
- Leaky tank $\dot q = -q + u$: $a_d = e^{-T_s}$, $b_d = 1 - e^{-T_s}$ (T_s=0.5: 0.607, 0.393).
- Cart $\dot p = v, \dot v = u$: $A_d = \left[\begin{smallmatrix} 1 & T_s \\ 0 & 1 \end{smallmatrix}\right]$, $B_d = \left[\begin{smallmatrix} T_s^2/2 \\ T_s \end{smallmatrix}\right]$ (Euler: $B_d = [0, T_s]^\top$, misses $T_s^2/2$).
- Cart+friction $\dot v = -cv + u$: $A_d = \left[\begin{smallmatrix} 1 & \beta \\ 0 & e^{-cT_s} \end{smallmatrix}\right]$, $B_d = \left[\begin{smallmatrix} (T_s - \beta)/c \\ \beta \end{smallmatrix}\right]$, $\beta = \frac{1 - e^{-cT_s}}{c}$.

W5 z-transform & z-plane

Four objects: $z_0 \in \mathbb{C}$ (number) → z_0^n (signal) → $H(z_0)$ (eigenvalue, a number) → $H(z)$ (function on ROC).

Modes $z^n = r^n e^{j\theta n}$: r = growth/decay envelope, θ = phase advance per sample. $r<1$ decay, $r=1$ sustained, $r>1$ growth; $z=1$: const., $z=-1$: $(-1)^n$.

$e^{j\Omega n}$ periodic ⟺ $\frac{\Omega}{2\pi} = \frac{m}{N} \in \mathbb{Q}$; N = period if $\frac{m}{N}$ irreducible.

Eigenfunction: $u[n] = z_0^n \Rightarrow y[n] = H(z_0) z_0^n$ with $H(z_0) = \sum_k h[k] z_0^{-k}$ (if it converges).

Definition $X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}$, $z \in$ ROC. $H(z) = \mathcal{Z}\{h\}$ = transfer function; $Y = HU$.

Properties

$\alpha x_1 + \beta x_2$	$\alpha X_1 + \beta X_2$
$x[n-k]$, $x[n+k]$	$z^{-k}X$, $z^k X$ (zero IC)
$(u*h)[n]$	$U(z)H(z)$
$a^n x[n]$	$X(z/a)$
$n\,x[n]$	$-z\frac{d}{dz}X(z)$
$x[-n]$	$X(1/z)$
$\sum_{k\le n} x[k]$	$\frac{z}{z-1}X(z)$
IVT (causal)	$x[0] = \lim_{z\to\infty} X(z)$
FVT (if exists)	$x[\infty] = \lim_{z\to 1}(z-1)X(z)$

Pairs (causal: ROC $|z| >$ largest pole)

$\delta[n]$ / $\delta[n-k]$	1 / z^{-k}
$s[n]$	$\frac{z}{z-1}$
$a^n s[n]$	$\frac{z}{z-a} = \frac{1}{1-az^{-1}}$, $\lvert z\rvert > \lvert a\rvert$
$-a^n s[-n-1]$	$\frac{z}{z-a}$, $\lvert z\rvert < \lvert a\rvert$ (left-sided)
$a^{n-1} s[n-1]$	$\frac{1}{z-a}$
$n\,a^n s[n]$	$\frac{az}{(z-a)^2}$ $\quad (ns[n]: \frac{z}{(z-1)^2})$
$(n+1)a^n s[n]$	$\frac{z^2}{(z-a)^2}$
$r^n\cos(\theta n)s[n]$	$\frac{z(z - r\cos\theta)}{z^2 - 2r\cos\theta\, z + r^2}$
$r^n\sin(\theta n)s[n]$	$\frac{r\sin\theta\, z}{z^2 - 2r\cos\theta\, z + r^2}$

! Expression alone is not unique: need ROC or causality.

ROC rules: never contains poles; right-sided (causal) ⇒ $|z| > \max|p_k|$; left-sided ⇒ $|z| < \min|p_k|$; two-sided ⇒ annulus; causal FIR ⇒ all $z\neq 0$. Stable (any LTI) ⟺ ROC ∋ unit circle.

Obtaining $H(z)$ **LCCDE** (delay $\to z^{-k}$):

$$H(z) = \frac{b_0 + \cdots + b_M z^{-M}}{1 + a_1 z^{-1} + \cdots + a_N z^{-N}}$$

then multiply num. & den. by $z^{\max(N,M)}$. Ex: $y[n] = \frac{1}{3}y[n-1] + u[n] - u[n-1] \Rightarrow H = \frac{z-1}{z-1/3}$.

State space: $zQ = AQ + BU \Rightarrow$

$$H(z) = C(zI - A)^{-1}B + D$$

with $(zI-A)^{-1} = \frac{\operatorname{adj}(zI-A)}{\det(zI-A)}$. Poles ⊆ eigenvalues of A (an eigenvalue may cancel with a zero). Ex: $(\frac{1}{2},1,\frac{1}{2},1)$: $\frac{1/2}{z-1/2} + 1 = \frac{z}{z-1/2}$.

Poles & zeros

$H(z) = b_0 z^{N-M}\frac{(z-z_1)\cdots(z-z_M)}{(z-p_1)\cdots(z-p_N)}$; zeros ∘ (num. roots), poles × (den. roots). $N>M$: zeros at 0; $M>N$: poles at 0.

Inverse via partial fractions Expand $\frac{H(z)}{z} = \frac{d}{z} + \sum_k \frac{c_k}{z-p_k}$,

$c_k = \left[(z-p_k)\frac{H(z)}{z}\right]_{z=p_k}$, $d = H(0)$:

$$H(z) = d + \sum_k \frac{c_k z}{z - p_k}$$

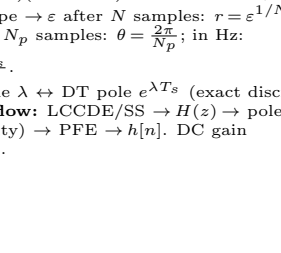
$$\Rightarrow\ h[n] = d\,\delta[n] + \sum_k c_k p_k^n\, s[n]$$

One mode p_k^n per pole; weights from numerator. Ex: $\frac{z^2}{(z-\frac{1}{2})(z-\frac{1}{4})} = \frac{2z}{z-\frac{1}{2}} - \frac{z}{z-\frac{1}{4}} \Rightarrow h = \left(2(\frac{1}{2})^n - (\frac{1}{4})^n\right)s[n]$.

Complex pair $p, \bar p$: $cp^n + \bar c\bar p^n = 2|c|\,r^n\cos(\theta n + \angle c)$.

Stability & pole locations **Causal LTI: BIBO stable ⟺ all poles strictly inside unit circle** ($|p_k|<1$) ⟺ ROC contains $|z|=1$. SS: all $|\lambda_i(A)|<1$ (sufficient). Zeros irrelevant for stability.

Mode of $p = re^{j\theta}$: $\propto r^n\cos(\theta n)$. $0<p<1$ monotone decay (faster as $r\to 0$); $-1<p<0$ alternating decay; $re^{\pm j\theta}$, $r<1$ decaying oscill.; $|p|=1$ sustained (tone), $p=1$ constant; $|p|>1$ diverges. r = decay rate, θ = osc. freq.



Pole placement Pair: $(z - re^{j\theta})(z - re^{-j\theta}) = z^2 - 2r\cos\theta\, z + r^2$.

Envelope → ε after N samples: $r = \varepsilon^{1/N}$. Period N_p samples: $\theta = \frac{2\pi}{N_p}$; in Hz: $f = \frac{\theta f_s}{2\pi}$.

CT pole $\lambda \leftrightarrow$ DT pole $e^{\lambda T_s}$ (exact discr.).

Workflow: LCCDE/SS → $H(z)$ → poles (stability) → PFE → $h[n]$. DC gain $= H(1)$.